

True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. The degree of precision of the following quadrature rule is 1:

$$\int_{-1}^1 f(x) \, dx \approx 2f(0).$$

2. The closed Newton–Cotes rule with $n + 1$ points is exact for all polynomials of degree up to n .

3. The trapezoidal approximation is always wrong if f is not a polynomial. That is to say, for every non-polynomial function f , it holds that

$$\int_{-1}^1 f(x) \, dx \neq f(-1) + f(1).$$

4. The degree of precision of the following quadrature rule is 1:

$$\int_{-1}^1 f(x) \, dx \approx f(1).$$

5. Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n + 1$ equidistant integration points. Then it holds that

$$\lim_{n \rightarrow \infty} |I[f] - I_n[f]| = 0, \quad I[f] := \int_a^b f(x) \, dx.$$

6. Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n + 1$ equidistant integration points. Then there exists C such that

$$\forall n \in \{1, 2, \dots\}, \quad |I[f] - I_n[f]| \leq \frac{C}{n^2}.$$

7. Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n + 1$ equidistant integration points. Then it holds that

$$\lim_{n \rightarrow \infty} n^2 |I[f] - I_n[f]| = 0.$$

8. There exist w_1 and w_2 such that the degree of precision of the following rule is 2:

$$\int_{-1}^1 f(x) \, dx \approx w_1 f(0) + w_2 f(1).$$

9. In Julia, the following code implements the composite trapezoidal rule with $n + 1$ points:

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    return h*sum(f, x[1:n] .+ h/2)
end
```

10. In Julia, the following code implements the composite Simpson rule with $n + 1$ points, where n is even:

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    result = 0.
    result += h/3 * f(x[1]) + h/3 * f(x[end])
    result += 4h/3 * sum(f, x[2:2:end-1])
    result += 2h/3 * sum(f, x[3:2:end-2])
    return result
end
```